

AI & Machine Learning for Civil Engineers

From engineering data to prediction, explainability and uncertainty, in eight hands-on sessions.

08

Live sessions

90 min

Per session

04 weeks

Course duration

Runs 100% on Google Colab

Nothing to install, works even on a Chromebook

Built for beginners

No prior Python or machine learning required

Live code-along on Civil Engineering datasets

Materials · Buildings & Energy · Transport · Environment · Geotechnics

Hands-on notebooks, cross-domain assignments & solutions

Plus certificate & recordings (till 2 weeks after the course)

INSTRUCTOR**Dr. Abdul Waris**

Research Associate

Universität Siegen (University of Siegen)

Former Prime Minister's Research Fellow

Ph.D. in Geotechnical Engineering, IIT Hyderabad

Research Excellence Award 2026, IIT Hyderabad

Invited talks at Oregon State University (USA) and a Faculty Development Programme at D.Y. Patil (Pune, India)

Resource person, five-day ML workshop at NIT Srinagar

Course outline**01 Python for Civil Engineers**

AI/ML basics, Colab, NumPy & pandas with engineering data

02 From Raw Data to ML-Ready

Explore, clean, visualize & split engineering data

03 Your First Prediction Model

Linear regression, error metrics & cross-validation

04 Trees & Ensemble Models

Decision Trees, Random Forest & boosting for nonlinear problems

05 Classification for Engineering Decisions

Logistic regression, confusion matrix, precision & recall

06 Artificial Neural Networks

Architecture, activations, training & overfitting

07 Explainable AI & Engineering Judgement

Feature importance, SHAP, and engineering interpretation

08 From Prediction to Probability

Monte Carlo, ML surrogates & probability of failure

Designed for Civil Engineers across disciplines — Materials, Structures, Geotechnics, Transportation, Water & Environment.

Early Bird offer till 14 October

Regular fees apply after 14 Oct

WHO SHOULD JOIN

B.Tech & M.Tech students / PhD scholars / Faculty / Practising engineers / Complete beginners to AI & ML

STARTS · LIMITED SEATS

**Saturday,
17 October 2026**

Scan to register · qudelastudios.com →

